Chapter 5

# Force and Motion — I

Newton's Laws of Motion: Complete Exam-Ready Study Guide

This guide walks through every concept, definition, law, derivation, figure, and worked relation from the chapter — Newton's First Law, force, inertial frames, mass, Newton's Second Law, the particular forces (gravity, weight, normal force, friction, tension), and Newton's Third Law — with all textbook Checkpoints answered and explained, a complete formula sheet, common mistakes to avoid, and a bank of practice questions with full solutions.

## Contents

# 1. Newton's First Law

**Historical background.** Before Newton, it was believed that a body needed a continuous 'force' to keep moving at constant velocity, and that a body's natural state was to be at rest. To keep something moving at constant velocity, it seemingly had to be continually pushed or pulled — otherwise it would 'naturally' stop.

This idea seems reasonable from daily experience: if you slide a puck across a wooden floor, it slows and stops. To keep it moving at constant velocity you must keep pushing it.

**The key insight — reducing friction.** Send the same puck sliding across ice instead, and it travels much farther before stopping, because ice is far less resistive than wood. Now imagine an idealized, perfectly slippery surface — a **frictionless surface** — over which the puck would hardly slow at all (this is closely approximated in the lab by an air table, where the puck rides on a thin film of air).

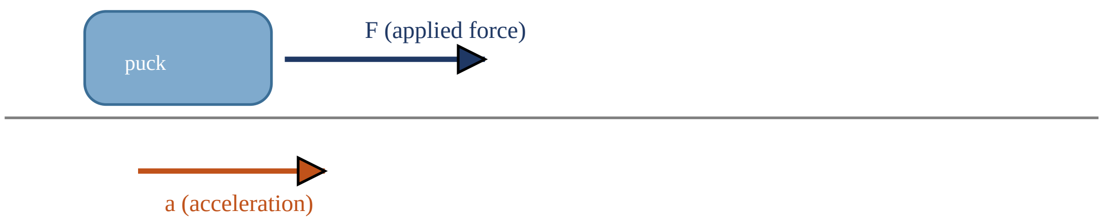



**Figure 5-1**

*A force F applied to the standard 1 kg body gives it an acceleration a. This is how the newton, the SI unit of force, is defined operationally.*

From the frictionless-surface thought experiment we conclude: **a body will keep moving with constant velocity if no force acts on it.** This is the seed of Newton's First Law.

> **★ Newton's First Law (initial statement)**
>
> If no force acts on a body, the body's velocity cannot change; that is, the body cannot accelerate.

In plain words: if a body is at rest, it stays at rest. If it is moving, it keeps moving with the same speed *and* the same direction (i.e., the same velocity vector) forever, unless something forces it to change.

**Refined version — the net force.** In real situations several forces often act on a body at once. What actually matters is not whether any individual force is zero, but whether the vector sum of all forces (the **net force**, F_net) is zero. Two people could each pull with 1 N and somehow get a net pull of 1.4 N (if pulling at an angle) — the world is predictable precisely because forces combine as vectors according to the **principle of superposition for forces**: a single force with the same magnitude and direction as the net of several forces produces the exact same effect as all of them acting together.

> **★ Newton's First Law (proper statement)**
>
> If no net force acts on a body (F_net = 0), the body's velocity cannot change; that is, the body cannot accelerate.

***Key takeaway for exams:*** *Constant velocity (including zero velocity, i.e. rest) ⇔ net force is zero. This works in both directions — if you know the net force is zero you can conclude the velocity is constant, and if you know the velocity is constant you can conclude the net force is zero. This reasoning is used constantly in equilibrium problems.*

## 2. Force — Units, Vectors, and Superposition

Before working numerical problems, we need to pin down exactly what a force *is*, how it is measured, and how multiple forces combine.

### Defining the Unit of Force

We define the force unit operationally, using the **standard kilogram** — a body whose mass is defined to be exactly 1 kg. Put this body on a horizontal, frictionless surface and pull it horizontally so that it accelerates at exactly 1 $m/s^2$. We *define* the force producing that acceleration to have a magnitude of 1 newton (symbol N).

If we pull with twice that force, the acceleration is found to double to 2 $m/s^2$ — i.e. force and acceleration are **directly proportional**. So if the 1 kg standard body has acceleration of magnitude a, the force producing it (in newtons) has magnitude numerically equal to a. This gives us a complete, workable definition of the force unit.

### Forces Are Vectors

Force has both magnitude *and* direction, so it is a vector quantity. When two or more forces act on a body, we find the **net force** (also called the **resultant force**) by adding them as vectors, following the standard rules of vector addition (head-to-tail addition, or component-by-component addition). A single force equal to this vector sum would produce exactly the same effect on the body as all the individual forces acting together — this is the **principle of superposition for forces**, and it is what makes everyday forces predictable.

Forces are usually written with an arrow over the symbol, such as F, and the net force is written F_net. When all the forces on a body act along a single axis, we can drop the vector notation and just use plus/minus signs to indicate direction along that axis.

### Checkpoint 1 — Vector Addition of Forces

The textbook shows six diagrams (a)-(f), each attempting to add two force vectors F1 and F2 tip-to-tail to form the resultant F_net. The question asks which of the six arrangements correctly represent vector addition.

**CHECKPOINT 1**

Which arrangement(s) correctly show the vector addition of F1 and F2 to form F_net?

**Answer & reasoning:** Correct vector addition requires placing the tail of the second vector at the head (tip) of the first vector (head-to-tail method), OR equivalently forming a parallelogram with F1 and F2 as adjacent sides and drawing the resultant as the diagonal from the common starting point. Diagrams in which the vectors are joined tail-to-tail without forming a proper parallelogram, or in which the 'resultant' is drawn as one of the original sides rather than the diagonal/head-to-tail closing vector, are incorrect. The parallelogram-style diagrams (where F1 and F2 originate from the same point and the resultant is the diagonal of the parallelogram they form) are the valid constructions — this is the general rule to apply to any such checkpoint: always check whether the diagram is a genuine head-to-tail chain or a true parallelogram with the resultant as the diagonal.

***Exam tip:*** *Vector addition problems like this test whether you can recognize the head-to-tail (polygon) method versus the parallelogram method, and can tell the resultant vector apart from the two original vectors in a diagram. Practice sketching both methods for two arbitrary vectors.*

# 3. Inertial Reference Frames

Newton's First Law is not true in *every* reference frame — it only holds in special frames called **inertial reference frames** (often shortened to inertial frames).

> **★ Definition**
>
> An inertial reference frame is one in which Newton's laws hold.

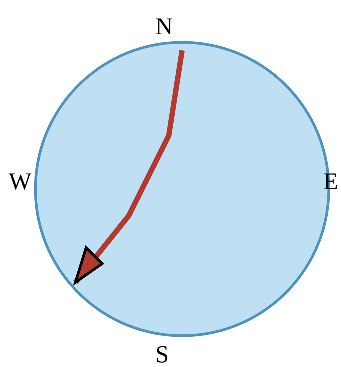
For most everyday problems we simply assume the ground is an inertial frame, which is valid provided we can neglect Earth's astronomical motions (such as its rotation).

## Why This Assumption Can Break Down: the Rotating-Earth Example

The assumption that the ground is inertial works well for short-range motion, such as a puck sliding along a short strip of frictionless ice. But consider a puck sent sliding along a very *long* ice strip extending outward from the North Pole.

• Viewed from a stationary frame out in space, the puck simply travels south in a straight line — Earth's rotation slides the ice underneath the puck, but the puck's own path is a straight line, exactly as Newton's First Law predicts (no net force acts on it, so it doesn't accelerate).

• Viewed from a point on the ground — i.e., a frame that rotates along with Earth — the puck's path is *not* straight. Because points on the ground move eastward faster the farther they are from the pole (greater eastward speed farther south), the ground underneath the puck keeps 'outrunning' it eastward as the puck slides south. From the ground observer's point of view, the puck appears to curve toward the west.




Path curves westward as seen from a rotating Earth (fictitious deflection)

**Figure 5-2**

*(a) Seen from a stationary point in space, the puck's path from the North Pole is a straight line south. (b) Seen from the rotating ground, the same puck appears to curve — an apparent westward deflection.*

This apparent deflection is **not** caused by any real force acting on the puck — it is purely an artifact of viewing the motion from a rotating (non-inertial) frame. If someone insisted on explaining the curve using Newton's laws by inventing a force to account for it, that invented force would be a **fictitious force** — a force that appears necessary only because the observer's frame is accelerating (in this case, rotating) rather than inertial.

A more everyday example of a fictitious force: in a car that is rapidly speeding up, you feel shoved back into the seat. It can feel as if a force is pushing you backward, but no such backward push actually exists — the car (and you, if strapped in) is accelerating forward under a real force, and the sensation is caused by observing the situation from the accelerating (non-inertial) reference frame of the car.

***Ground rule used throughout this book:*** *Unless stated otherwise, we assume the ground is an inertial frame, and that all measured forces and accelerations are made relative to the ground. If measurements are instead made from an accelerating vehicle (a noninertial frame), the results can look surprising and must be interpreted with care — this is exactly the situation you must watch for in elevator and vehicle-acceleration problems.*

# 4. Mass

Everyday experience tells us that applying the *same* force to different bodies (say, a baseball and a bowling ball) produces different accelerations — the more massive object accelerates less. Newton's insight, made precise, is that acceleration is **inversely proportional** to mass (not, say, inversely proportional to the square of the mass).

## Deriving the Mass Relation

Push on the standard body (mass exactly m0 = 1 kg) with a force of magnitude 1 N; it accelerates with magnitude a0 = 1 m/s². Now push on some other body X with that *same* force, and suppose it accelerates with magnitude aX = 0.25 m/s². We make the (correct) assumption that, for the same applied force, the ratio of masses equals the inverse ratio of accelerations:

$$m_X / m_0 = a_0 / a_X$$

Solving for the unknown mass gives:

$$m_X = m_0 (a_0 / a_X) = (1.0 \text{ kg})(1.0 \text{ m/s}^2 / 0.25 \text{ m/s}^2) = 4.0 \text{ kg}$$

## Checking Consistency

This definition of mass is only useful if it gives the *same* answer no matter what force we happen to use for the comparison. As a check: apply an 8.0 N force first to the standard body (getting acceleration 8.0 m/s²) and then to body X (getting acceleration 2.0 m/s²):

$$m_X = m_0 (a_0 / a_X) = (1.0 \text{ kg})(8.0 \text{ m/s}^2 / 2.0 \text{ m/s}^2) = 4.0 \text{ kg}$$

We get the same mass, 4.0 kg, both times. This confirms the procedure is **consistent** and therefore a legitimate, usable way to define and measure mass.

## What Mass Actually Is

These results show that mass is an **intrinsic characteristic** of a body — a property that comes automatically with the body's existence, independent of where it is or what forces act on it. Mass is also a **scalar** quantity (no direction).

It is natural to wonder what mass 'really' is in an intuitive sense — is it size? weight? density? The answer is **no** to all of these, although people often confuse mass with these other properties. The most precise statement we can make is definitional:

> ***Mass is the characteristic of a body that relates a force on the body to the resulting acceleration.*** *There is no more familiar, everyday definition than this — you only get a physical 'feel' for mass when you try to accelerate a body yourself, such as kicking a baseball versus kicking a bowling ball.*

# 5. Newton's Second Law

All of the definitions, experiments, and observations about force and mass are summarized in one compact, central statement of mechanics:

**★ Newton's Second Law**

The net force on a body is equal to the product of the body's mass and its acceleration.

$$F_{net} = m\,a \qquad \text{(Newton's Second Law, Eq. 5-1)}$$

## Rule 1 — Identify the Body

This equation is the single most important tool for nearly every homework and exam problem in this chapter, but it must be used with care. You must always be certain *which* body you are applying it to. F_net must be the vector sum of *all* the forces acting on *that* body — and only forces acting on that body. You must never include forces acting on some other body that happens to be involved in the situation.

Example: if you are in a rugby scrum, the net force on *you* is the vector sum of every push and pull acting on your own body — it does **not** include any push or pull that one other player exerts on a different player. Every time you tackle a force problem, your very first step should be to clearly state which body you are applying Newton's second law to.

## Rule 2 — Separate the Axes

Like any vector equation, Eq. 5-1 is equivalent to three independent component equations, one for each axis of an x, y, z coordinate system:

$$F_{net,x} = m\,a_x \quad F_{net,y} = m\,a_y \quad F_{net,z} = m\,a_z \qquad \text{(Eq. 5-2)}$$

Each of these equations relates the net force component along *one* axis to the acceleration component along that *same* axis, and is completely unrelated to force components along any other axis.

***General rule (memorize this):*** *The acceleration component along a given axis is caused only by the sum of the force components along that same axis, and not by force components along any other axis. This is the single most exam-tested idea in this chapter — it is exactly why we resolve every force into x- and y-components before applying F_net = ma along each axis separately.*

## Forces in Equilibrium

Equation 5-1 tells us that if the net force on a body is zero, the body's acceleration a = 0. If the body is at rest, it stays at rest; if it is moving, it continues at constant velocity. In such cases the forces on the body are said to **balance** one another, and are also said to **cancel** one another.

***Careful with the word 'cancel':*** *Canceling forces does NOT mean the forces cease to exist (canceling forces is not like canceling a dinner reservation!). The individual forces still fully act on the body — they simply add to zero as vectors, so together they cannot change the body's velocity.*

## Units

For SI units, Eq. 5-1 tells us directly how the newton relates to the kilogram, meter, and second:

$$1\ \mathrm{N} = (1\ \mathrm{kg})(1\ \mathrm{m/s^2}) = 1\ \mathrm{kg \cdot m/s^2} \quad \text{(Eq. 5-3)}$$

**Table 5-1. Units in Newton's Second Law**

| System | Force | Mass | Acceleration |
|---|---|---|---|
| SI | newton (N) | kilogram (kg) | $m/s^2$ |
| CGS | dyne | gram (g) | $cm/s^2$ |
| British | pound (lb) | slug | $ft/s^2$ |

*Conversion notes: 1 dyne = 1 $g \cdot cm/s^2$. 1 lb = 1 $slug \cdot ft/s^2$.*

# 6. Free-Body Diagrams

To solve problems using Newton's second law, we typically draw a **free-body diagram**, in which the *only* body shown is the one for which we are summing forces. To save space and keep the focus on forces, the body is usually represented by a single dot. Each force acting on the body is drawn as a vector arrow with its tail anchored on that dot. A coordinate system is usually included, and if the body accelerates, its acceleration is also drawn (labeled clearly as an acceleration, distinct from any force). This entire procedure exists to focus your attention sharply on the body of interest and prevent you from accidentally including forces on other bodies.

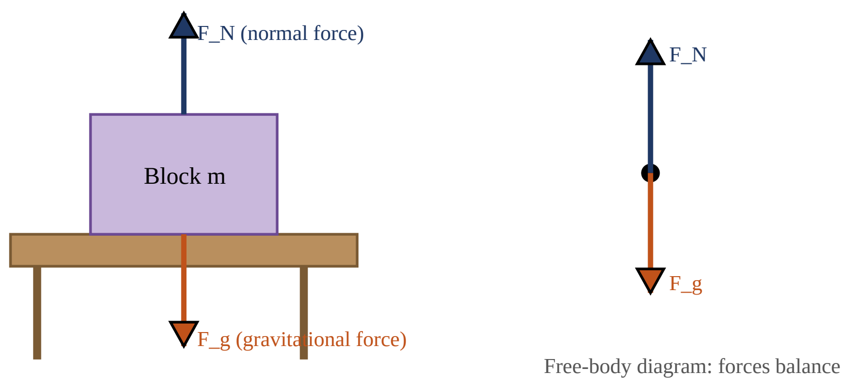



**Figure 5-7**

*(a) A block of mass m rests on a table. Gravity F_g pulls the block down; the table pushes back up with normal force F_N. (b) The free-body diagram for the block: the body is reduced to a dot, and only the two forces acting on the block itself are drawn. Because the block doesn't accelerate vertically, the two forces are equal and opposite — they balance.*

### Worked Example — Block on a Table

Forces F_g and F_N are the only two forces on the block, and both act along the vertical. Choosing a positive-upward y axis, Newton's second law (F_net,y = m a_y) for the block reads:

$$F_N - F_g = m\, a_y$$

Substituting F_g = mg (derived in Section 7 below):

$$F_N = m\, g + m\, a_y = m\,(g + a_y) \quad \text{(Eq. 5-13)}$$

This holds for *any* vertical acceleration of the table and block together — for instance if they are inside an accelerating elevator. (Caution: the sign for g is already built into the formula, but a_y itself can be positive or negative depending on whether the acceleration is upward or downward.) If the table and block are not accelerating relative to the ground, a_y = 0 and the normal force simply equals the weight:

$$F_N = m\, g \quad \text{(Eq. 5-14, non-accelerating case)}$$

### Checkpoint 3 — Normal Force in an Elevator

### CHECKPOINT 3

In the block-on-table setup, is the magnitude of the normal force $F_N$ greater than, less than, or equal to mg if the block and table are in an elevator moving upward (a) at constant speed and (b) at increasing speed?

**Answer & reasoning:** (a) At **constant speed**, the acceleration $a_y = 0$, so from Eq. 5-13, $F_N = m(g + 0) = mg$. The normal force is **equal to** mg — constant velocity (even though it's upward and nonzero) still means zero acceleration. (b) At **increasing speed** while moving upward, the elevator (and block) is accelerating *upward*, so $a_y$ is positive. From Eq. 5-13, $F_N = m(g + a_y) > mg$. The normal force is **greater than** mg — this is exactly the extra 'heavy' feeling you sense when an elevator starts moving upward.

# 7. Some Particular Forces

## 7.1 The Gravitational Force

A **gravitational force F_g** on a body is a certain type of pull that is directed toward a second body. In introductory chapters we don't discuss the deeper nature of this force, and we usually consider the second body to be Earth. So 'the' gravitational force on a body usually means a force pulling it straight down, toward Earth's center. We assume the ground is an inertial frame.

### Free Fall

Suppose a body of mass m is in free fall with the free-fall acceleration of magnitude g. If we neglect air effects, the *only* force acting on the body is the gravitational force F_g. Placing a vertical y axis along the body's path with positive direction upward, Newton's second law (F_net,y = m a_y) becomes, in this situation:

$$-F_g = m(-g) \quad \Rightarrow \quad F_g = m\,g \qquad \text{(Eq. 5-8)}$$

In words: the magnitude of the gravitational force is equal to the product mg.

### At Rest — The Force Doesn't Disappear

This same gravitational force, with exactly the same magnitude mg, still acts on a body even when the body is *not* in free fall — for instance while it rests on a table or moves across it. (For the gravitational force to disappear, Earth itself would have to disappear.) In vector form:

$$F_g = -F_g \blacksquare = -mg \blacksquare = m\,g\blacksquare \qquad \text{(Eq. 5-9)}$$

where ■ is the unit vector pointing upward along the y axis, away from the ground, and g■ is the free-fall acceleration written as a vector, pointing downward.

## 7.2 Weight

The **weight W** of a body is defined as the magnitude of the net force required to prevent the body from falling freely, as measured by someone standing on the ground.

Example: to hold a ball at rest in your hand while standing on the ground, you must supply an upward force that balances the gravitational force on the ball. If that gravitational force has magnitude 2.0 N, your upward force must also be 2.0 N, and the ball's weight W is therefore 2.0 N. We also say the ball *weighs* 2.0 N. A ball with a weight of 3.0 N would require a greater 3.0 N force from you to hold — we say this second ball is *heavier* than the first, because the gravitational force you must balance is larger.

### General Derivation of W = mg

Consider a body with acceleration a = 0 relative to the ground (an inertial frame). Two forces act on it: a downward gravitational force F_g, and a balancing upward force of magnitude W. Newton's second law along a positive-upward y axis (F_net,y = m a_y) gives, in this situation:

$$W - F_g = m(0) \qquad \text{(Eq. 5-10)}$$

**$W = F_g$ (weight, with ground as inertial frame) (Eq. 5-11)**

**★ Weight**

The weight W of a body is equal to the magnitude $F_g$ of the gravitational force on the body.

Substituting mg for $F_g$ (from Eq. 5-8) gives the familiar working formula:

$$W = m g \quad \text{(weight)} \quad \text{(Eq. 5-12)}$$

## Weighing a Body

To *weigh* a body means to measure its weight. One method uses an **equal-arm balance**: place the unknown body on one pan, then add reference bodies of known mass to the other pan until the device balances (i.e., the gravitational forces on the two sides match). Since the gravitational forces match, the masses match too, so we now know the mass of the body — and, if we know g at that location, we can find its weight from $W = mg$.

A second method uses a **spring scale**: the body stretches a spring, moving a pointer along a calibrated scale. If the scale is marked in weight units, its reading directly gives the weight of whatever sits on the pan. If instead it's marked in mass units, the reading only correctly gives the object's mass if the value of g at the location where the scale is being *used* is the same as the value of g where the scale was originally *calibrated*.

The weight of a body must be measured while the body is **not accelerating vertically** relative to the ground. If you weigh yourself on a scale inside an accelerating elevator, the reading will differ from your true weight because of that acceleration — such a reading is called an **apparent weight**.

***Critical distinction — weight is NOT mass:*** *Weight is the magnitude of a force and is related to mass by $W = mg$. If you move a body to a location where g has a different value, the body's mass (an intrinsic property) does not change, but its weight does. Example: a bowling ball with mass 7.2 kg weighs 71 N on Earth but only about 12 N on the Moon, because the Moon's free-fall acceleration is only about 1.6 $m/s^2$ (versus roughly 9.8 $m/s^2$ on Earth) — the mass, 7.2 kg, is exactly the same in both places.*

## 7.3 The Normal Force

If you stand on a mattress, Earth pulls you downward, yet you remain stationary. The reason: the mattress deforms downward under your weight, and in doing so it pushes back up on you. A seemingly rigid floor does exactly the same thing, just imperceptibly (it is compressed, bent, or buckled ever so slightly) — enough people standing on a floor not resting directly on the ground could, in principle, break it.

The push you feel from the mattress or floor is called a **normal force, F_N**. The name comes from the mathematical meaning of 'normal': perpendicular. The force on you from a surface, such as the floor, is always directed **perpendicular to that surface**.

**★ Normal Force**

When a body presses against a surface, the surface (even a seemingly rigid one) deforms and pushes back on the body with a normal force F_N that is perpendicular to the surface.

The general relation for the magnitude of a normal force acting on a body resting on a horizontal surface, allowing for possible vertical acceleration a_y (see the free-body-diagram derivation in Section 6), is:

$$F_N = m(g + a_y) \quad \text{(Eq. 5-13);} \quad \text{reduces to } F_N = mg \text{ when } a_y = 0 \text{ (Eq. 5-14)}$$

## 7.4 Friction

If we slide, or even just attempt to slide, a body over a surface, the motion is resisted by a bonding between the body and the surface. This resistance is treated as a single force, f, called either the **frictional force** or simply **friction**. Friction is directed along the surface, **opposite to the direction of the intended (or attempted) motion**.

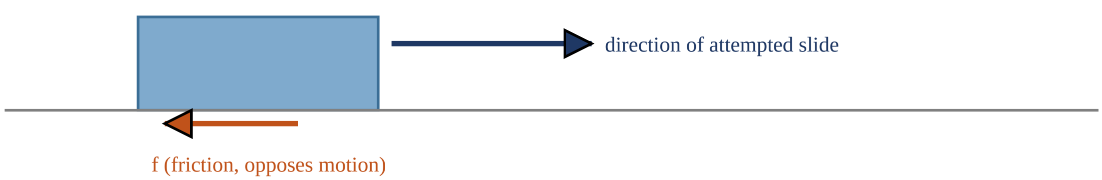



**Figure 5-8**

*A frictional force f opposes the attempted slide of a body over a surface — note that f points opposite to the direction the body is being pushed/pulled, not necessarily opposite to any existing motion.*

Sometimes, to simplify a problem, friction is assumed to be negligible — in that case the surface (or even the body itself) is said to be **frictionless**. (This chapter treats friction only qualitatively here; a full quantitative treatment with coefficients of friction comes in the next chapter.)

## 7.5 Tension

When a cord (or rope, cable, or similar object) is attached to a body and pulled taut, the cord pulls on the body with a force T that is directed **away from the body, along the cord**. This is often called a **tension force**, because the cord is said to be 'in tension' (or 'under tension') — meaning it is being pulled taut. The **tension in the cord** is simply the magnitude T of the force it exerts. For example, if the force from the cord on a body has magnitude 50 N, then the tension in that cord is 50 N.

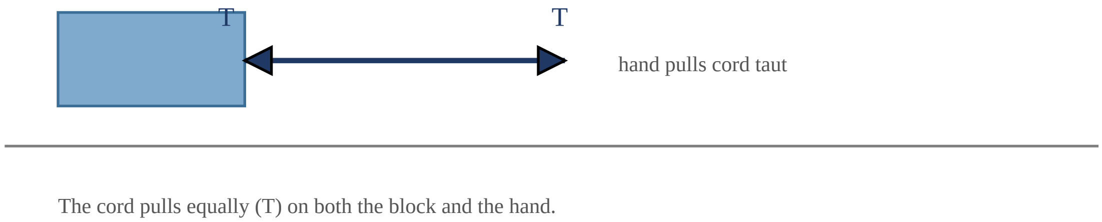



**Figure 5-9**

*A cord pulled taut is under tension. If its mass is negligible, the cord pulls on the body at one end and the hand (or support) at the other end with the same force magnitude T — even if the cord runs around a massless, frictionless pulley.*

A cord is often idealized as **massless** (its mass is negligible compared to the bodies it connects) and **unstretchable**. Under these idealizations the cord exists purely as a connection between two bodies: it pulls on both bodies with the *same* force magnitude T, even if the bodies and cord are accelerating, and even if the cord runs around a massless, frictionless pulley. Such a pulley has negligible mass compared to the bodies involved and negligible friction opposing rotation on its axle.

**Special case — cord wrapped around a pulley:** if the cord wraps *halfway* around a pulley (as when it changes the direction of pull by 180°), the *net* force on the pulley from the cord has magnitude 2T (the two straight segments of cord each pull on the pulley with force T, and — because they point in the same net direction after the wrap — their effects add).

## Checkpoint 4 — Tension While Accelerating

### CHECKPOINT 4

A suspended body weighs 75 N. Is the tension T equal to, greater than, or less than 75 N when the body is moving upward (a) at constant speed, (b) at increasing speed, and (c) at decreasing speed?

**Answer & reasoning:** Take up as positive. Newton's second law for the body: $T - W = m a_y$, so $T = W + m a_y$. (a) **Constant speed** ⇒ $a_y = 0$ ⇒ $T = W =$ **75 N (equal to)**. (b) **Increasing speed while moving upward** ⇒ acceleration is upward ($a_y > 0$) ⇒ $T = W + m a_y > 75$ N, so T is **greater than 75 N**. (c) **Decreasing speed while moving upward** means the body is decelerating, so its acceleration points downward ($a_y < 0$) ⇒ $T = W + m a_y < 75$ N, so T is **less than 75 N**. General pattern to remember: tension increases above the static weight whenever the net acceleration of the hanging body is upward, and decreases below the static weight whenever the net acceleration is downward — regardless of which direction the body currently happens to be moving.

## 8. Newton's Third Law

Two bodies are said to **interact** when they push or pull on each other — that is, when a force acts on each body due to the other body. Example: suppose you lean a book B against a crate C. The book and crate interact: there is a horizontal force F_BC on the book from (or 'due to') the crate, and a horizontal force F_CB on the crate from (or 'due to') the book.

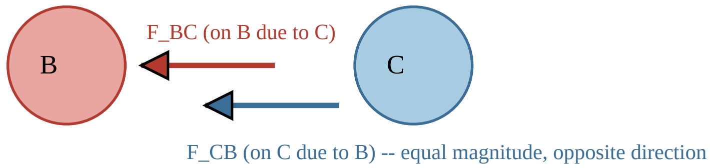



**Figure 5-10**

*(a) Book B leans against crate C. (b) The forces F_BC (on the book, from the crate) and F_CB (on the crate, from the book) have the same magnitude and act in opposite directions.*

> **★ Newton's Third Law**
>
> When two bodies interact, the forces on the bodies from each other are always equal in magnitude and opposite in direction.

For the book and crate, this law can be written as the scalar relation:

$$F_{BC} = F_{CB} \quad \text{(equal magnitudes)}$$

or, more completely, as the vector relation:

$$F_{BC} = -F_{CB} \quad \text{(equal magnitudes, opposite directions)} \quad \text{(Eq. 5-15)}$$

where the minus sign signals that the two forces point in opposite directions. The two forces between two interacting bodies are together called a **third-law force pair**. This holds whenever any two bodies interact in any situation — the book and crate could be moving, or even accelerating, and the third law would still hold exactly.

### Worked Example — Cantaloupe, Table, and Earth

Consider a cantaloupe lying on a table that itself stands on Earth. The cantaloupe interacts with both the table and with Earth, so there are three bodies whose various interactions we must carefully sort out.

**Step 1 — forces acting on the cantaloupe alone.** Two forces act on the cantaloupe itself: F_CT, the normal force on the cantaloupe from the table, and F_CE, the gravitational force on the cantaloupe due to Earth. Are these two a third-law force pair? **No** — even though they happen to be equal and opposite (which is why the cantaloupe doesn't accelerate), a third-law pair must consist of forces on *two different, interacting bodies*. F_CT and F_CE are both forces on the *same single body* (the cantaloupe), so they cannot be a third-law pair — they are simply two forces that happen to balance.

**Step 2 — the cantaloupe-Earth interaction.** To find an actual third-law pair, we must look at the interaction *between* two different bodies. Earth pulls on the cantaloupe with gravitational force F_CE, and by Newton's third law, the cantaloupe simultaneously pulls back on Earth with gravitational force F_EC. These forces act on two different, interacting bodies (the cantaloupe and Earth), so they *are* a genuine third-law pair:

$$F_{CE} = -F_{EC} \quad \text{(cantaloupe–Earth interaction)}$$

**Step 3 — the cantaloupe-table interaction.** Similarly, in the cantaloupe-table interaction, the force on the cantaloupe from the table is F_CT, and conversely the force on the table from the cantaloupe is F_TC. These, too, act on two different interacting bodies (the cantaloupe and the table), so they are also a genuine third-law pair:

$$F_{CT} = -F_{TC} \quad \text{(cantaloupe–table interaction)}$$

***The single most important exam distinction in this whole topic:*** *A third-law force pair always consists of two forces acting on two different bodies — one force on each body, due to the other. Two forces acting on the same body (like normal force and gravity on a resting object) can be equal and opposite by coincidence of equilibrium, but they are* ***never*** *a third-law pair, because a third-law pair is defined by the interaction between two bodies, not by numerical equality alone. A fast way to check: third-law pairs always have matching subscripts reversed, e.g. F_AB and F_BA — never F_AB paired with some unrelated F_AC.*

# 9. Master Formula Sheet

| Concept | Equation | Notes |
|---|---|---|
| **Newton's First Law** | $F_{net} = 0$ ⇒ $a = 0$ | Body at rest stays at rest; body moving keeps constant velocity |
| **Newton's Second Law** | $F_{net} = m a$ | Vector equation; identify the body; sum only forces on that body |
| **Second Law by axis** | $F_{net,x} = ma_x$<br>$F_{net,y} = ma_y$<br>$F_{net,z} = ma_z$ | Each axis is independent of the others |
| **SI force unit** | 1 N = 1 kg·m/$s^2$ | Defined via 1 kg standard body accelerating at 1 m/$s^2$ |
| **Gravitational force** | $F_g = m g$ | Acts on a body whether falling, at rest, or moving |
| **Weight** | $W = F_g = m g$ | Weight = magnitude of gravitational force; NOT the same as mass |
| **Normal force (general)** | $F_N = m(g + a_y)$ | $a_y$ is the body's vertical acceleration (elevator, etc.) |
| **Normal force (no accel.)** | $F_N = m g$ | Special case: $a_y = 0$ |
| **Friction** | f opposes attempted/actual sliding | Directed along the surface, opposite intended motion |
| **Tension (ideal cord)** | T is same throughout a massless, frictionless-pulley cord | Pulls both connected bodies toward each other along the cord |
| **Force at pulley (180° wrap)** | $F_{pulley} = 2T$ | Net force on pulley from a cord wrapped halfway around it |
| **Newton's Third Law** | $F_{AB} = -F_{BA}$ | Equal magnitude, opposite direction; acts on two different bodies |

## Units Quick Reference (Table 5-1)

| System | Force | Mass | Acceleration |
|---|---|---|---|
| SI | newton (N) | kilogram (kg) | m/$s^2$ |
| CGS | dyne | gram (g) | cm/$s^2$ |
| British | pound (lb) | slug | ft/$s^2$ |

## 10. Common Mistakes & Exam Tips

### 1. Mixing forces on different bodies

When applying $F_{net} = ma$, students often accidentally add in a force acting on a different object. Fix: always draw a free-body diagram for exactly one body first, and only include arrows for forces acting directly on that body.

### 2. Confusing weight with mass

Weight ($W = mg$, measured in newtons) is a force and depends on location (value of $g$). Mass (measured in kilograms) is intrinsic and does not change with location. Never say a body 'weighs 5 kg' — that is a mass, not a weight.

### 3. Misidentifying third-law pairs

A third-law pair is never two forces on the same body (like normal force and gravity on a resting block) — even though they may be equal and opposite. A genuine third-law pair always involves one force on each of two different, mutually interacting bodies.

### 4. Forgetting that 'cancel' does not mean 'disappear'

When forces are said to balance/cancel, both forces still physically act on the body; they simply sum to a zero net vector. This matters when a follow-up question asks about the reaction force to one of the two.

### 5. Applying Newton's laws in a non-inertial (accelerating/rotating) frame without adjustment

Newton's first and second laws hold as stated only in inertial frames. In an accelerating elevator or a rotating frame, apparent (fictitious) effects appear that are not caused by any real force — recognize when a problem is set in a non-inertial frame.

### 6. Sign errors with acceleration direction in $F_N$ or $T$ problems

Always define a clear positive direction (usually upward) at the start of a vertical-motion problem, and be consistent about the sign of $a_y$ throughout — $a_y$ is positive for upward acceleration and negative for downward acceleration, regardless of the direction of the body's current velocity.

### 7. Treating cords/pulleys as realistic instead of idealized

Unless a problem specifically says otherwise, assume cords are massless and unstretchable, and pulleys are massless and frictionless. Under these assumptions the tension is the same throughout the cord.

### 8. Forgetting to separate the second law into components

For 2-D problems (inclined planes, angled ropes, etc.), always resolve every force into x- and y-components first, then apply $F_{net,x} = ma_x$ and $F_{net,y} = ma_y$ as two separate equations.

## 11. Practice Question Bank (With Full Solutions)

### Q1. Conceptual — First Law

A hockey puck slides across a perfectly frictionless surface at constant velocity. What is the net force acting on it? Explain using Newton's First Law.

**Solution:** By Newton's First Law, since the puck moves at constant velocity (no change in speed or direction), its acceleration is zero. Because $F_{net} = ma$, and $a = 0$, the net force on the puck must be exactly **zero**. This does not mean no forces act on it at all — gravity and the normal force from the ice may both be present — it means whatever forces do act sum to zero as vectors.

### Q2. Numerical — Second Law, single axis

A 5.0 kg block on a frictionless horizontal surface is pushed by a horizontal force of 15 N. Find its acceleration.

**Solution:** Apply Newton's second law along the horizontal (x) axis: $F_{net,x} = m a_x$. Since the surface is frictionless and the push is the only horizontal force, $F_{net,x} = 15$ N. $a_x = F_{net,x} / m = 15 \text{ N} / 5.0 \text{ kg} =$ **3.0 m/s²**, directed in the direction of the push.

### Q3. Numerical — Mass from Acceleration Comparison

A standard 1.0 kg body accelerates at 2.0 m/s² under a certain force. The same force applied to an unknown body Y produces an acceleration of 0.50 m/s². Find the mass of Y.

**Solution:** Using $m_Y/m_0 = a_0/a_Y$: $m_Y = m_0 (a_0/a_Y) = (1.0 \text{ kg})(2.0 \text{ m/s}^2 / 0.50 \text{ m/s}^2) = (1.0 \text{ kg})(4.0) =$ **4.0 kg**.

### Q4. Numerical — Weight vs. Mass

A astronaut has a mass of 80 kg. Find her weight (a) on Earth, where $g = 9.8$ m/s², and (b) on the Moon, where $g = 1.6$ m/s². Has her mass changed between the two locations?

**Solution:** Using $W = mg$: (a) On Earth: $W = (80 \text{ kg})(9.8 \text{ m/s}^2) =$ **784 N** ≈ **$7.8 \times 10^2$ N**. (b) On the Moon: $W = (80 \text{ kg})(1.6 \text{ m/s}^2) =$ **128 N** ≈ **$1.3 \times 10^2$ N**. Her mass is **unchanged** at 80 kg in both locations — mass is intrinsic and location-independent; only her weight (a force) changes because g is different.

### Q5. Numerical — Normal Force in an Accelerating Elevator

A 60 kg person stands on a scale inside an elevator. Find the scale reading (the normal force) if the elevator accelerates (a) upward at 2.0 m/s², and (b) downward at 2.0 m/s². Use $g = 9.8$ m/s².

**Solution:** Use $F_N = m(g + a_y)$, taking up as positive. (a) Upward acceleration: $a_y = +2.0$ m/s². $F_N = (60 \text{ kg})(9.8 + 2.0) \text{ m/s}^2 = (60 \text{ kg})(11.8 \text{ m/s}^2) =$ **708 N** (heavier than static weight of 588 N). (b) Downward acceleration: $a_y = –2.0$ m/s². $F_N = (60 \text{ kg})(9.8 – 2.0) \text{ m/s}^2 = (60 \text{ kg})(7.8 \text{ m/s}^2) =$ **468 N** (lighter than static weight).

## Q6. Numerical — Tension for a Body Accelerating Upward

A 10 kg object is lifted straight up by a cord with an acceleration of 3.0 $m/s^2$. Find the tension in the cord. Use $g = 9.8$ $m/s^2$.

**Solution:** Take up as positive. Newton's second law on the object: $T – mg = m a_y$, so $T = m(g + a_y)$ = (10 kg)(9.8 + 3.0) $m/s^2$ = (10 kg)(12.8 $m/s^2$) = **128 N**. Since the object accelerates upward, tension exceeds the static weight of 98 N, consistent with the Checkpoint 4 pattern.

## Q7. Conceptual — Third Law Identification

A book rests on a table. Identify the third-law force pair partner for (a) the gravitational force of Earth on the book, and (b) the normal force of the table on the book.

**Solution:** (a) The third-law partner of 'Earth pulls book down' is **the book pulling Earth up** with equal magnitude (the book's own gravitational pull on Earth) — NOT the normal force from the table. (b) The third-law partner of 'table pushes book up (normal force)' is **the book pushing down on the table** with equal magnitude — NOT the gravitational force on the book. (This tests the key distinction from Section 8: forces on the same body, like gravity and the normal force on the book, are never a third-law pair even though they're equal and opposite here.)

## Q8. Numerical — Two-Body System (connected bodies)

Two blocks, $m1 = 2.0$ kg and $m2 = 3.0$ kg, are connected by a massless cord and pulled across a frictionless horizontal surface by a horizontal force of 20 N applied to m2 (which is in front, pulling m1 behind it). Find (a) the acceleration of the system and (b) the tension in the connecting cord.

**Solution:** (a) Treat the two blocks as one composite system of total mass $M = m1 + m2 = 5.0$ kg. The only external horizontal force is the applied 20 N. $a = F_{net} / M$ = 20 N / 5.0 kg = **4.0 $m/s^2$**. (b) Now isolate m1 alone (the trailing block); the only horizontal force on it is the cord tension T pulling it forward: $T = m1\, a$ = (2.0 kg)(4.0 $m/s^2$) = **8.0 N**. (Check: for m2, $F_{net}$ = 20 N – T = 20 – 8.0 = 12 N, and a = 12 N / 3.0 kg = 4.0 $m/s^2$ — consistent with part (a).)

## Q9. Conceptual — Inertial Frames

A ball is placed on the floor of a car that is accelerating forward. Relative to someone sitting in the car, the ball rolls backward even though nothing appears to push it. Explain this observation using the concept of inertial vs. noninertial frames.

**Solution:** The car is an accelerating (noninertial) reference frame. In the truly inertial frame of the ground outside, the ball actually obeys Newton's First Law: with no horizontal force on it, it simply stays at rest (or keeps moving at constant velocity) while the car accelerates forward underneath it. To the observer inside the accelerating car, however, it looks as though the ball spontaneously rolls backward. This apparent backward motion is not caused by any real force — it is a **fictitious effect** that arises purely from viewing the situation from a noninertial (accelerating) frame, exactly analogous to the rotating-Earth puck-deflection example.

### Q10. Numerical — Pulley With 180° Wrap

A cord under 40 N of tension is wrapped halfway around a fixed, massless, frictionless pulley, changing the direction of pull by 180° (as in Fig. 5-9c). Find the net force the cord exerts on the pulley.

**Solution:** For a cord wrapped halfway (180°) around a pulley, the net force on the pulley from the cord has magnitude 2T. Net force = 2(40 N) = **80 N**, directed along the bisector of the angle between the two cord segments (here, straight down toward whatever the cord supports).

## 12. Quick-Recall Summary Card

Use this page as a last-minute, night-before-the-exam refresher. If you can explain every line below in your own words, you are ready.

- **First Law:** $F_{net} = 0 \Rightarrow$ velocity constant (includes rest). Only true in inertial frames.
- **Inertial frame:** a reference frame in which Newton's laws hold; the ground is usually assumed inertial. A rotating or accelerating frame is noninertial and produces fictitious effects.
- **Mass:** intrinsic, scalar property relating force to acceleration ($m = F/a$ for a fixed force); does not depend on location.
- **Second Law:** $F_{net} = m a$ (vector equation); split into $F_{net,x} = ma_x$, $F_{net,y} = ma_y$, $F_{net,z} = ma_z$; each axis independent of the others. Always identify the one body you're analyzing.
- **Free-body diagram:** show only forces acting directly on the one chosen body, as vectors anchored at a dot representing that body.
- **Gravitational force:** $F_g = mg$, always present (not just in free fall).
- **Weight:** $W = F_g = mg$ — a force, in newtons, that depends on g at the location; NOT the same as mass.
- **Normal force:** $F_N$ perpendicular to the surface; $F_N = m(g + a_y)$ generally, reducing to $F_N = mg$ when there's no vertical acceleration.
- **Friction:** f acts along the surface, opposing attempted or actual sliding.
- **Tension:** T is the same throughout an ideal (massless, unstretchable) cord, even around a massless, frictionless pulley; net force on a pulley from a 180° wrap is 2T.
- **Third Law:** $F_{AB} = -F_{BA}$, always between two different, interacting bodies — never two forces on the same body.

> ***Final exam strategy:*** *For every problem, (1) identify the single body of interest, (2) draw its free-body diagram, (3) choose axes and resolve every force into components, (4) apply $F_{net,x} = ma_x$ and $F_{net,y} = ma_y$ separately, (5) solve algebraically before plugging in numbers, and (6) sanity-check signs and units at the end. This five/six-step routine handles the overwhelming majority of Newton's-law exam problems.*